

Kartik Chauhan

Full Stack Engineer | AI Engineer | Agentic AI Developer
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Professional Summary

Full-Stack Software Engineer with 3+ years of experience building scalable web applications and AI-powered systems using the MERN stack and modern AI frameworks. Experienced in high-performance applications, REST APIs, payment systems, real-time architectures, scalable backend architecture, cloud deployment, vector databases, AI automation systems, RAG pipelines, LLM integrations, LangChain, and LangGraph. Improved application performance by up to 40% through database optimization and scalable frontend engineering, with strong hands-on experience in production-grade applications, performance optimization, and full-stack development.

Technical Skills

Programming Languages: JavaScript, TypeScript, Python, C++, SQL, HTML5, CSS3

Frontend: React.js, Redux, Tailwind CSS, Bootstrap, Vite, Next.js

Backend and APIs: Node.js, Express.js, REST APIs, Socket.io, JWT Authentication

AI and Agentic AI: LLMs, LangChain, LangGraph, RAG, Retrieval-Augmented Generation, AI Agents, Prompt Engineering, Vector Databases, AI Workflow Automation, OpenAI APIs, Gemini API, Hugging Face, LangFuse, Multi-Agent Systems, Embeddings, Semantic Search

Databases: MongoDB, MySQL, Pinecone, ChromaDB

Cloud and DevOps: AWS, Docker, Kubernetes, Render, GitHub Actions

Tools and Platforms: Git, GitHub, Postman, Swagger, Jira, VS Code

Work Experience

Associate Delivery Level 3 | SmartData Enterprises Inc. | Dehradun, Uttarakhand, India | Jan 2026 – Present

- Working on scalable healthcare sector applications with a focus on backend architecture, AI workflows, and enterprise-grade system development.
- Developed AI-powered workflows for report generation, audio report processing, automated summarization, and intelligent admin dashboards.
- Built scalable backend APIs and optimized data-processing workflows improving system efficiency and reducing manual operations.
- Contributed to optimization initiatives that improved API response performance and reduced backend processing overhead by approximately 35%.
- Collaborated with cross-functional teams using Agile methodologies and modern development workflows.
- Improved application performance, maintainability, and scalability across multiple production-grade healthcare projects.

Technology Senior Executive | Pearl Organisation | Dehradun, Uttarakhand, India | May 2025 – Jan 2026

- Managed and delivered 20+ projects including eCommerce platforms, vehicle booking systems, tour booking applications, and business management solutions.
- Handled multiple client projects simultaneously while ensuring timely delivery, production stability, and business-critical workflow support.
- Developed scalable web applications and backend services, improving frontend responsiveness through optimized rendering and reusable UI architecture.
- Integrated secure payment gateway systems and scalable backend workflows for high-traffic applications while supporting debugging, testing, deployment, and production operations.

Software Developer | Codementees | Bengaluru, Karnataka, India | Sep 2024 – May 2025

- Developed a Full-Stack Mentorship Platform using MERN stack, connecting hundreds of mentees with experienced tech mentors.
- Built scalable backend APIs supporting authentication, real-time communication, and content management features, including Socket.io messaging that increased engagement by 20%.
- Optimized MongoDB queries, aggregation pipelines, and indexing strategies, improving database performance by 40% and reducing API response times significantly.
- Improved frontend maintainability and performance by building reusable React components, eliminating repetitive code, developing SEO-friendly pages, and applying Next.js server-side rendering best practices.

MERN Stack Developer | Quitepaper | Bikaner, Rajasthan, India | Jul 2023 – Aug 2024

- Designed and developed a scalable Library Management System with seat booking and CRM functionality used by multiple organizations.
- Engineered backend workflows and optimized query execution to improve booking efficiency and reduce system

latency.

- Built an automated seat allocation algorithm reducing booking conflicts by 30% and implemented real-time notifications to improve platform responsiveness.
- Integrated secure payment gateway systems capable of handling 1000+ concurrent users using Socket.io, TypeScript, MongoDB, and scalable backend architectures.

Software Engineer Intern | Canvas Craft Media | Sahibzada Ajit Singh Nagar, Punjab, India | Jan 2023 – May 2023

- Worked on frontend development using React.js and modern JavaScript frameworks.
- Assisted in developing responsive UI components and integrating REST APIs.
- Collaborated with senior developers on debugging, testing, feature implementation, full-stack development, and Agile workflows.

Projects

AutoNodeAPI – Open Source NPM Package

- Developed an open-source Node.js package that automates CRUD API generation for Mongoose models.
- Integrated Swagger documentation for automatic API references and improved developer onboarding.
- Designed configurable routing and middleware support for scalable backend architecture.
- Reduced backend boilerplate code and improved API development speed.

AI Storytelling and Agentic Workflow Platform

- Built AI-powered workflows for automated story generation, image generation, subtitles, and audio creation using LLMs and AI APIs.
- Automated multi-step content pipelines reducing manual content generation efforts by approximately 70%.
- Developed agentic AI pipelines using LangChain and LangGraph for multi-step workflow orchestration.
- Implemented RAG-based context retrieval systems using vector databases to improve AI response quality.
- Integrated OpenAI and Gemini APIs for conversational AI and intelligent automation systems.

Jobgram

- Developed a MERN-based job portal platform for recruiters and job seekers.
- Implemented advanced filtering, authentication, and recruiter job management workflows.
- Built responsive UI components and scalable backend APIs.
- Tech Stack: React.js, Node.js, Express.js, MongoDB, Vercel.

TripNest – Smart Tour and Vehicle Booking Platform

- Developed a scalable full-stack travel and booking platform supporting tour booking, vehicle reservations, event booking, and smart travel planning.
- Implemented intelligent tour plan suggestions and smart route optimization features to improve user travel experience.
- Built a real-time thread and discussion system allowing travelers and organizers to communicate seamlessly.
- Integrated secure payment gateway systems capable of handling high concurrent booking traffic.
- Developed advanced search filters, booking workflows, admin dashboards, and responsive UI architecture.
- Optimized frontend performance using reusable React components and scalable backend APIs.
- Tech Stack: React.js, Next.js, Node.js, Express.js, MongoDB, Socket.io, Stripe.

Education

Bachelor of Technology in Computer Science and Engineering | Punjab Technical University | 2019 – 2023